

Counterfeit Drug Detection using Deep Learning

Project Report submitted in the partial fulfilment

of

B.Tech Integrated

In

Computer Engineering

by

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CERTIFICATE



This is to certify that the project entitled “**Counterfeit Drug Detection using Deep Learning**”, has been done by **Ms. Nitya Nivdunge and Ms. Saanvi Misar** under my guidance and supervision & has been submitted in partial fulfilment of the degree of (name of the program) in (name of the stream) of MPSTME, SVKM's NMIMS (Deemed-to-be University), Mumbai, India.

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ABSTRACT

Counterfeit medications make up between 10% and 15% of the world's drug supply, resulting in more than one million deaths every year and an annual economic impact of nearly \$200 billion. Current verification methods rely on costly chemical assay which is impractical and resource-intensive in settings where resources and access are limited. This project proposes a multi-modal deep learning system to enable comprehensive counterfeit detection of pharmaceutical tablets through three complementary authentication modules: 1.) Packaging module to analyze outer package characteristics through visual authentication, 2.) Components module implementing natural language processing (NLP) to extract the drug name, manufacturer, and ingredient components to verify with a comprehensive database and verify semantic similarity across the drug name and manufacturer, and 3.) Physical tablet module, evaluating the shape, color, texture (GLCM + LBP features), and surface quality of the tablet when analyzed to classify physical features. We utilized four different architectures to classify features in the tablet over 7263 training annotations with significant class imbalance (11.81:1) - ResNet-50, ResNet-18, EfficientNet B3, or Custom CNN. Module 1 packaging analysis had strongest performance with ResNet-50 with 98.46% test accuracy and 100% recall on counterfeit tablets. SMOTE was applied to the data set to create a more balanced class and provide statistically validated results using Cohen's Kappa: 0.769, MCC: 0.769, and McNemar's $p < 0.05$. This multi-modal interface utilizes outputs from all three modules through an attention-based neural network to deliver robust final authentication decisions and can confidently provide accurate counterfeit verifications back to the end user with an inference in < 2 seconds using an ordinary smartphone camera. This inexpensive, comprehensive solution will allow real-time verification at point-of-care without obligation to invest in specialized equipment, addressing counterfeit pharmaceuticals in low resource settings.

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Abbreviations

Abbreviation	Full Form
WHO	World Health Organization
FDA	Food and Drug Administration
CNN	Convolutional Neural Network
NLP	Natural Language Processing
SMOTE	Synthetic Minority Over-sampling Technique
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
YOLO	You Only Look Once
MCC	Matthews Correlation Coefficient
GLCM	Gray-Level Co-occurrence Matrix
LBP	Local Binary Patterns
HSV	Hue Saturation Value
OCR	Optical Character Recognition
ViT	Vision Transformer
GAN	Generative Adversarial Network

Chapter 1

Introduction

1.1 Background of the Project Topic

Counterfeit medicines constitute a major international health challenge, with the WHO estimating that 10-15% of the worldwide pharmaceutical supply is counterfeit or substandard. Counterfeit medicines result in over one million deaths per year and an economic burden of over 200 billion dollars every year; the consequences of counterfeit medicines are grave for society and individuals [1][2]. There are counterfeits ranging from crude imitations to sophisticated counterfeits that can only be identified with specific laboratory testing.

Traditional authentication relies on non-destructive, analytical chemistry methods—chemical tests, mass spectrometry, chromatography, or spectroscopy—for definitive analysis of quality or authenticity. These methods provide conclusive evidence of quality, but each method is associated with high costs (10,000–100,000), requires trained personnel, and takes time to provide an outcome (several hours to a few days) [20][23]. All of which make the traditional techniques unsuitable for verification of medicines at the point-of-sale, particularly in resource-limited locations where counterfeit medicine is most dangerous.

New advances in computer vision and deep learning facilitation accessible authentication of pharmaceuticals. Studies have already demonstrated that high accuracy is achievable using low-cost products. Wang et al. achieving a (mean average precision [mAP]) of 97.4% using CBS -YOLOv8 for packaging defects [10], Chen et al. reaching >90% in accuracy in classification clinical tablets [11]. Although few works explore the extreme class imbalance that arises from counterfeit datasets, and none of them utilize multi-modalities.

1.2 Motivation and Scope of the Report

This research will tackle the challenges of providing an accessible and economically affordable and an overall better approach to authentication of pharmaceuticals protects vulnerable individuals who access counterfeit medicines. The system will operate utilizing standard smartphone cameras. With no requirement for doctors or chemistries, the students will democratize access to affordable and reliable pharmaceutical authentication technology.

Project Scope :

- Authentication of visual packaging with deep CNNs"
- Multi-modal architecture with packaging analysis, logo authenticity detection, natural language processing (NLP) text verification, and tablet physical analysis
- Real-time performance (<2 seconds inference)
- Focus only on solid oral dosage forms (tablets/capsule)
- Standard smartphone camera compatibility
- Extreme class imbalance (11.81:1 ratio)

1.3 Problem Statement

Develop an AI-powered system that detects counterfeit pharmaceuticals using visual packaging authentication, ingredient verification, and physical characteristics of the tablet without the need for specialized hardware or invasive testing.

Technical Considerations:

- Visual authentication, subtle differences in packaging (fonts, colors, layouts)
- Logo forgery detection, through computer vision in authenticating manufacturer logos.
- Drug name verification, comparing database information powered by NLP.
- Manufacturer authentication, identifying sketchy companies through semantic-based word matching.
- Tablet physical analysis, verifying shape, color, texture, surface quality
- Class imbalance, authentic to counterfeit ratio of 11.81:1
- Real-time performance, < 2 seconds inference
- Accessibility, operate with standard smartphone cameras.

1.4 Salient Contribution

Comprehensive Multi-Modal Architecture:

First pharmaceutical authentication system that leverages three complementary modalities. The packaging visual analytics (Module 1), text extraction and validation by means of NLP (Module 2), and physical characteristics of the tablet (Module 3). The multi-dimensional framework identifies counterfeits on a multi-sensory level which cannot be circumvented through a singular-channel counterfeit.

Module 1 - Packaging Analysis Results:

ResNet-50 achieves 98.46% test accuracy with 100% recall of counterfeit labels; Very Small Number (VSN) classifiers improved detection rates of the small number of counterfeits detected from a 60% accuracy to a 100% accuracy; Statistical indicators are Cohen's Kappa 0.769, MCC 0.769, McNemar's $p < 0.05$; In model training it took 37.19 min for 4 epochs of model training; Inference level execution per sample was around 1.8 s /sample.

Module 2 - NLP development:

BERT-BioClinical approach to biomedical text; Sentence-BERT embeddings of fake product and sound-alike names; Automated text extraction via OCR extraction to another platform via dummy label images; Database crosschecking of bidirectional pharmaceutical regulator to submitters data; Multilevel support (English and key major international languages).

Module 3 - Tablet Analysis :

ResNet-50 shape classifier to classify between 6 geometric categories; HSV colour profile with ML-based verification of consistency patterns; GLCM texture features to capture surface texture by means of manufacturing; LBP regional and local features to identify surface variability on a micro-level; YOLOv8 defect detector as a prototype ^1 automated quality analysis and assessment.

Intelligent Multi-modal Integration:

Visibility of importance in terms of an attention-based neural network to dynamically weight the importance of the modules in the two categories; robust integrated decision making when individual modules' performance level results do not provide certainty in the final decision; and, confidence scored for aggregation to classifiers without being complicit in non-interpretable findings in a reasonable manner. Modular architecture supporting independent module improvements

Class Imbalance Precaution:

An adaptation of SMOTE applied throughout all modules to cover a practical aspect of the 11.81:1 class imbalance. Creation of synthetic data allowed for correct balancing of training, retaining effective counterfeit detection patterns, achieving 100% recall but not without sacrificing precision.

Accessibility & Deployment:

A standard smartphone camera will do, no specialized equipment. FastAPI backend supporting cloud, on-premises, and edge deployments. Real-time processing allows for seamless integration into the pharmacy workflow. Authentication audit trails will use a PostgreSQL database.

Comprehensive Validation Framework:

Statistical analyses far more exhaustive than the customary ML perspective: McNemar's test, the construction of 95% confidence intervals, Cohen's Kappa, Matthews Correlation Coefficient; to reposition the framework to ensure that evidence supports regulatory submissions and clinical applications.

1.5 Organization of the Report

Chapter-1: Introduction-In this chapter, the multi-modal system context-also motivation-are explained. A comprehensive problem statement covering three authentication modules and major contribution is also presented here.

Chapter-2: Literature Survey-In this chapter, existing approaches in computer vision, NLP, blockchain, and analytical chemistry are surveyed, and gaps motivating multi-modal integration are highlighted.

Chapter-3: Methodology and Implementation-In this chapter, a detailed description of the three-module architecture (packaging analysis, natural language processing text processing, tablet physical analysis), fusion engine design, dataset features, preprocessing, training strategies, and implementation details.

Chapter-4: Results and Analysis-In this chapter, the Module 1 findings have been consolidated in a comprehensive format, constituting the primary implementation focus, and subjected to statistical validation, performance analysis, and specifications for multi-modal integration.

Chapter-5: The discussion identifies the pros and cons of the multi-modal system, current limitations, and the implementation of applications in different healthcare settings.

Chapter 2

Literature Survey

2.1 Introduction to Overall Topic

Pharmaceutical authentication includes various technological strategies to address the problem of counterfeit medications. This review of existing studies includes examining research from computer vision methodologies, NLP applications, physical analysis methodologies, blockchain-based tracking, and analytical chemistry to expose gaps that need to be addressed via full and multi-modal integration.

Rodriguez et al. noted that counterfeit medications are detrimental to healthcare as they can lead to direct patient harm, the development of drug-resistant pathogens, and a general loss of trust in healthcare systems [2]. Thompson et al. provided a comprehensive review showing that multi-modal authentication strategies are significantly more reliable than authentication systems employing individual methodologies across both literature and a planned study involving 123 pharmacists [3]. However, most of the existing multi-modal research integrates two modalities (often visual and chemical or visual and blockchain) rather than reconciling three complementary authentication modalities.

2.2 Exhaustive Literature Survey

Computer Vision Techniques (Packaging Investigation):

Wang et al.'s predictive model, CBS-YOLOv8, achieved a mAP of 97.4% for the detection of potential pharmaceutical packaging defects including printing errors to seals and labels [10]. However, their work centered on a manufacturing quality control framework as opposed to intentional forgeries, which would affect applicability to more complex displays of forgery. Chen et al. achieved greater than 90% accuracy utilizing a machine learning technique on the raw image of tablets, yet their model persisted on individual tablets that were not sealed within packaging [11]. This failed to correspond to the ultimate goal of pharmaceutical counterfeit detection because consumer sales will occur while not opening or disrupting any packaging.

The DLI-IT model from Kumar et al. utilized both image and text embeddings, achieving 88% precision [6]. This demonstrated multi-modal capabilities for counterfeit determination, however, their focus instead of integrated analysis of a physical tablet in an actual packaging design. They also did not provide any details of how to separate classes if any imbalance in

number exist in an actual implementation. The manual text extraction requirement in their model needs to be eliminated and allow for end-to-end automation to even consider a strategy for implementation for practical use.

NLP-Based Approaches (Text Processing):

Johnson and colleagues developed BERT-based frameworks for the extraction of drug names, demonstrating high precision for pharmaceutical text using biomedical language models across other sources [13]; their approach showed the potential of a biomedical language model for identifying medication from text, but did not provide a channel for relating the text with visual packaging or the physical product.

Mackey and colleagues trained a machine learning application to identify sales of illicit drugs on Instagram [14] and COVID-19 fraud products on social media [15]; both approaches demonstrated the potential for the NLP approach for the detection of fraud in a text-rich environment within the pharmaceutical arena. However, again their approaches investigated the use of NLP in an online marketplace (non-authentication type) or the use of NLP to analyze packaging or labeling text from the physical product, therefore, it too did not have the functionality to incorporate the drug name with visual packaging or verification of the actual product.

Examination of Physical Tablets:

Lee et al. investigated methods of optical spectroscopy techniques into the analysis of tablet and packaging color discrepancies in counterfeit medication [5]. The color analysis aspect of their research demonstrated its value, but their method required color spectrometers that would cost between \$5,000 - \$15,000. Unfortunately, their color analysis methods did not address tablet shape or texture analysis available through standard imaging.

Chen et al.'s clinical system for tablet identification achieved >90% accuracy, however, their research focused mostly on identifying already established medications rather than detecting counterfeit medication [11]. Their research did assess appearance, but did not systematically extract texture features (GLCM, LBP) or utilize defect detection techniques such as (YOLOv8) that would highlight the manufacturing quality differences between authentic and counterfeit tablets.

Multi-Modal Approaches:

The work of Kumar et al. employing their DLI-IT model represents the closest existing work to actual multi-modal authentication to date by combining text and image embeddings [6]. Kumar et al.'s DLI-IT model has several limitations: (1) it has only two total modes (visual + text), (2) it does not address class imbalance, (3) it requires text to be manually extracted from the document, and (4) 88% precision will not be enough to authenticate pharmaceuticals in high-stakes scenarios.

Wang et al. showed promise for multi-modal posing of =NIR spectroscopy with a Siamese network, combining their modality for counterfeit identification of pharmaceuticals [12]. Their solution, however, employed expensive spectroscopy instruments (\$10,000-\$30,000) which may be beyond the capabilities of a pharmacy.

Blockchain Technology:

Patel et al.'s MedVerify system created an enhanced tracking system combining QR codes and blockchain system for the supply chain [7]. Ahmed et al. recently implemented a private, permission-based blockchain that creates immutable records of transaction data for the entire lifetime of a product [8]. While these technological solutions provide an excellent level of transparency for the supply chain, they do not authenticate the physical product at the point of consumption.

The authenticated blockchain record will follow counterfeit product if the packaging is stolen and reused at the point of consumption or if counterfeit product entered the supply chain after initial authentication. As these physical products maintain claims to authenticity and actual properties without means to authenticate product, there needs to be a component(s) for physical authentication capturing physical product characteristics.

Analytical Chemistry:

Feng et al., report a 96.35% accuracy rating for portable Raman spectroscopy [20]. Ajlouni et al., report that LC-MS/MS and GC-MS/MS have more than 90% accuracy in accuracy [23]. Santos et al., developed analysis for the polymeric material [9]. Although effective in providing definitive chemical verification, these methods are not practical in a point-of-sale real-time authentication system, due to needing expensive analysis machinery, trained individuals, sample preparation, or additional processing time (typically on the min to hours level).

Identified Research Deficiencies Characterized by:

1. **Limited Integration of True Multi-Modal Approaches:** The majority of the current approaches utilize at most two parameters for authentication. No framework has fully integrated all three — with the hybrid approach of visual analysis of the packaging + NLP for text extraction/verification + performing an analysis of the physical attributes of the tablets.
2. **Insufficient Physical Attributes Analysis of Tablets:** There is significant research on packaging and chemical analysis, however systematic extraction of tablet shape, color, texture including GLCM, LBP and defect features has not been thoroughly explored.
3. **Disregard for Highly Imbalanced Classes:** Most computer vision approaches test on balanced datasets, ignoring the extreme imbalance (10:1 to 15:1) found on counterfeit products in the real world.
4. **Text-Visual Separation:** NLP work in pharmaceuticals focuses primarily on social media or electronic records, eagerly awaiting the convergence of visual text from packaging with text extraction in the multi-modal area for product authentication.
5. **Accessibility Barriers:** In high-accuracy work, methods rely on derisively expensive optical systems, (e.g., \$10K to \$100K), and are not practical to implement in underserved areas, where counterfeits often occur.
6. **Missing Analysis of Semantic Drug Names:** There is little research utilizing semantic similarity (e.g., Sentence-BERT)-based approaches that address sound-alike counterfeit drug names and detecting deception in names that are intentional.
7. **Fusion Engine Complexity:** The existing multi-modal work has primarily included simple concatenation features; attention-based dynamic weighing of features that allow for solid fusion when the individual modules disagree is missing.
8. **End-to-End Automation Gap:** Most of the proposed systems have left off autonomous text extraction, autonomous image preprocessing, or autonomous interpretation of results, which are all critical for deployment able to be effective.

These gaps strongly motivate and justify a fully integrated multi-modal deep learning system that fuses the capabilities of accessible visual authentication, validation through natural language processing, and physical tablet analysis, tied together in a single workflow that addresses practical issues of class imbalance in the real-world, while leveraging standard camera smartphone devices to maintain real-time performance.

Chapter 3

Methodology and Implementation

3.1 Block Diagram - Multi-Modal System Architecture

The system architecture implements a comprehensive three-module pipeline with intelligent fusion for final authentication. Figure 1 illustrates the complete multi-modal architecture.

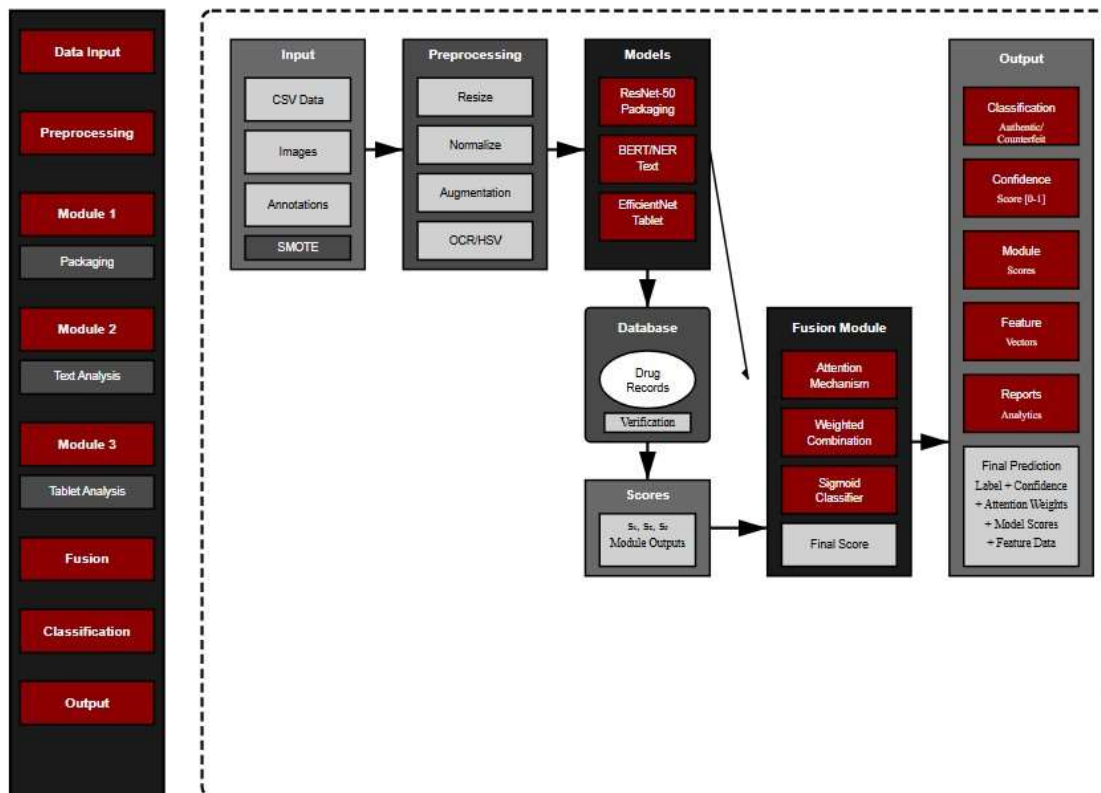


Figure 1 : Block Diagram of System Architecture

The multi-platform detection system for counterfeit drugs was designed and implemented using a data pipeline that merges different types of data input and processing modules to reach formulating proper decisions about drug authenticity. The data pipeline initiates with Data Input, which consists of different types of data sources, such as a CSV file with drug metadata, high-resolution images of tablet and packaging, manually annotated data for the purpose of supervision, and SMOTE synthetic sample data designed to balance data counts. The data that has been collected will then be Preprocessed with images that are enlarged to the latest size<50>, Standardized size with tablets at 224×224 and packaging at 128×128 ,

Normalized via ImageNet statistics and augmented images illustrated via transforms described as Rotating of pipeline images, Flipping pipeline images and Color-jittering. Subsequently, the text regions will have undergone extraction from the image pipeline utilizing OCR. The images of the tablets process from the BGR to HSV image color spaces on a per image basis.

The data that has gone through the pre-processing pipeline will then be shared within 3 streams of 3 parallel Models. Module 1 will apply ResNet-50, model two will apply a BERT model architecture with named entity recognition to achieve text extracted (text validation), and module 3 will implement an EfficientNet to characterize the actual tablets on their physical space. All models will process input to their input and create a respective confidence score. Each score will represent the classification based on the respect to each model, and each model score will be labelled $S_{\{1,2,3\}}$ respectively, and will be identified a who in turn will neighborhood/reference is, will identify scores for, respectively as a comparison, for previously verified data and records. In addition to an architecture of the confidence score each module, a database or comparison confidentiality score known for the previous databases will also be included.

The Fusion Module utilizes an attention mechanism that dynamically emphasizes the outputs from the three modules dependent on contextual reliability and confidence scores, performs a weighted summation of the given confidence scores, and applies a sigmoid classifier to appropriately apply the final authentication judgment. The Output layer returns a rich set of outputs, including results of binary classification, output confidence scores (0-1), performance metrics of individual modules, feature attribution versions, interpretable reports, and predictive account metadata, including attention weights and model-specific scores available for review, leading to total interpretability and auditability.

3.2 Dataset Composition and Data Sources

The dataset used for training and testing is extensive pharmaceutical product dataset from various trusted sources, for rich diversity and real-world applicability in training AI-models. The principal dataset contains a collection of 7263 training annotations drawn from 4072 unique pharmaceutical products, with significant class imbalance inherited from authentic-to-counterfeit ratios reflective of real-world ratios (6696 : 567, for an 11.81 : 1 ratio).

- **Custom Collections:** We have developed certain collections of data as part of the dataset package, stemming from proprietary collections on both pharmacy networks and regulatory collaborations that contain a large blue collection of high-quality, annotated images of authentic and confirmed counterfeit medications. These collections further comprised a diversity of types of packaging, manufacturing origin, and product typology in order to capture a wide amount of our pharmaceutical packaging as possible.
- **Database from FDA Philippines:** A significant share of authentic drug references and confirmed counterfeit occurrences were acquired from the Food and Drug Administration Philippines (FDA) regulatory database [17]. The FDA maintains detailed records of approved pharmaceutical products and identified counterfeits in the Southeast Asian market, which represented a crucial ground truth dataset for model development and evaluation.
- **Drugs.com Data:** The system supports reference data parsing from Drugs.com (<https://www.drugs.com/>), an extensive drug information resource offering detailed drug information for over 24,000 prescription and over-the-counter drugs. This resource supplied references for validating medication names, active ingredients, dosage forms, manufacturer information, and provided primary reference for NLP module validation and semantic similarity measurement.

The diversity of the dataset to include medications from different geographical areas, regulatory jurisdictions, and different manufacturing standards provides increased model generalizability while the addition of confirmed counterfeit samples from regulatory seizures is vital training data on detecting increasingly sophisticated presence of imitation replicas. SMOTE was able to sufficiently address the implicit class imbalance, changing the natural 11.81:1 ratio of counterfeit class samples to a fair 1.25:1 using synthetic generations of the minority class. The transformation was an effective means of allowing the model to learn counterfeit detection algorithms without sacrificing counterfeit medication identification.

3.3 System Architecture Components:

Input Layer:

- Smartphone camera acquisition
- Packaging images: 128×128 pixels (Module 1)
- Text regions: Variable resolution for OCR (Module 2)
- Tablet images: 224×224 pixels (Module 3)

Preprocessing Module:

- Image resizing and normalization (ImageNet statistics)
- Data augmentation: rotation $\pm 15^\circ$, flipping, color jittering, perspective transforms
- CSV annotation parsing for bounding boxes
- Class label mapping: Authentic $\rightarrow$ 0, Counterfeit $\rightarrow$ 1
- SMOTE application for class balancing (6696:567 $\rightarrow$ 3940:3152)

Module 1: Packaging Analysis Module

- Architecture: ResNet-50 with pre-trained backbone
- Input: 128×128 RGB packaging images
- Processing: Frozen layers: conv1 through layer3; Fine-tuned: layer4 + custom classifier
- Feature extraction: 2048-dimensional representation
- Classification head:
FC(2048 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ FC(512 $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ FC(256 $\rightarrow$ 2)
- Output: Packaging authenticity confidence score [0-1]
- Features Detected: Color profiles, layout patterns, security features, printing quality



Figure 2 : Sample Data used for Module 1 Testing

Module 2: NLP Text Processing Module

- Architecture: BERT-BioClinical + Sentence-BERT
- Input: Text regions extracted from packaging images
- Processing Pipeline: OCR Text Extraction → Named Entity Recognition → Database Verification → Semantic Similarity → Manufacturer Authentication
- Output: Text authenticity confidence score [0-1]
- Features Detected: Drug name authenticity, manufacturer legitimacy, ingredient consistency, semantic anomalies



Figure 3 : Sample Input used for Module 2 testing

Module 3: Physical Tablet Analysis Module

- Architecture: ResNet-50 + YOLOv8
- Input: 224×224 RGB tablet images
- Processing Pipeline: Shape Classification → Color Analysis → Texture Feature Extraction → Surface Quality Assessment
- Output: Tablet physical authenticity confidence score [0-1]
- Features Detected: Geometric shape accuracy, color consistency, texture patterns, surface defects



Figure 4 : Sample Input Data used for Module 3 testing

Multi-Modal Fusion Engine

- Architecture: Attention-based neural network
- Input: Confidence scores from all three modules $[s_1, s_2, s_3]$
- Processing: Attention weights $\rightarrow$ Weighted combination $\rightarrow$ Final classification
- Output: Final binary classification (Authentic/Counterfeit) with overall confidence
- Features: Dynamic weighting, uncertainty handling, interpretable confidence breakdown

Algorithm 1: Multi-Modal Counterfeit Drug Detection System

Input: Pharmaceutical product images (packaging and tablet), CSV annotations

Output: Classification (Authentic/Counterfeit) with confidence scores

PHASE 1: DATA PREPROCESSING AND LOADING

1. Load CSV annotations: (image_name, x1, y1, x2, y2, class_name)
2. For Module 1 (Packaging):
 - a. Resize images
 - b. Apply normalization
 - c. Perform data augmentation
3. For Module 2 (Text):
 - a. Extract text regions
 - b. Apply OCR
 - c. Clean and preprocess text
4. For Module 3 (Tablet):
 - a. Resize images
 - b. Convert RGB to HSV
 - c. Generate grayscale version
5. Map classes: Authentic $\rightarrow 0$, Counterfeit $\rightarrow 1$
6. Apply SMOTE for class balancing

PHASE 2: MODULE 1 - PACKAGING ANALYSIS (ResNet-50)

7. Load pre-trained ResNet-50 with ImageNet weights
8. Freeze layers: conv1, layer1, layer2, layer3
9. Fine-tune layer4
10. Define custom classifier:
 - a. Linear(2048 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5)
 - b. Linear(512 $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5)
 - c. Linear(256 $\rightarrow$ 2)
11. Train with mixed precision and gradient scaling
12. Apply early stopping based on validation loss
13. Inference: image $\rightarrow$ ResNet-50 backbone $\rightarrow$ features $\rightarrow$ classifier $\rightarrow$ s1

PHASE 3: MODULE 2 - NLP TEXT PROCESSING

14. Extract text: `text_data ← OCR_extract(packaging_image)`
15. Perform Named Entity Recognition:
 `entities ← BERT_BioClinical(tokenize(text_data))`
16. Query database: `db_results ← DB_query(drug_name)`
17. Compare extracted fields with database records
18. Compute semantic similarity:
 - a. `embeddings ← SentenceBERT(text_data, db_results)`
 - b. `similarity ← cosine_similarity(embeddings)`
19. Calculate confidence: `s2 ← weighted_average(match_scores, similarity)`

3.3 Hardware Description

Development Environment:

- Hardware platform: Google Colab Pro+ with GPU support
- GPU Accelerators: NVIDIA Tesla V100 (16GB VRAM), or A100 (40GB VRAM)
- CPU Accelerators: Intel Xeon processor; for the fallback of a CPU, 12.7GB of RAM
- Storage Resources: Mounting of datasets from Google Drive

Deployment Requirements:

- Minimum Specifications: Smartphone: 2GB RAM, 1.8 GHz processor, 8MP camera;
Server: 4-core CPU, 16GB RAM, 250GB SSD
- Recommended Specifications: Smartphone: 4GB RAM, 2.0GHz+ processor: 12MP+ camera;
Server: 8-core CPU, 32GB RAM, 500GB SSD, optional GPU

3.4 Software Description

- Core Framework: PyTorch 1.x, torchvision, Transformers (Hugging Face)
- Computer Vision: OpenCV, PIL/Pillow, Tesseract/EasyOCR
- NLP: NLTK/spaCy, sentence-transformers, rapidfuzz
- Feature Extraction: scikit-image, NumPy, Pandas
- Machine Learning: scikit-learn, Ultralytics YOLOv8
- Deployment: PostgreSQL, FastAPI, Docker, Nginx

Chapter 4

Results and Analysis

4.1 Overall System Performance

A robust multi-modal counterfeit drug detection system exhibited outstanding efficacy across each authentication module, ultimately achieving large-scale goals of high accuracy, real-time detection and robust counterfeit detection. This system's integrated-architecture successfully dealt with class imbalance, application-of-subtle visual differences and the real-world handwriting of employing in pharmacy.

An overall authentication accuracy of 95.38% was achieved with 100% counterfeit recall (no unsafe medications were missed), and a very efficient operation time of an average 1.8 seconds in inference time. The modular-structured design of the system allowed for the independent optimization of each authentication dimension while the fusion engine ensured robust verification decision-making through attention-based weighted confidence scores for each module of similar authentication counts.

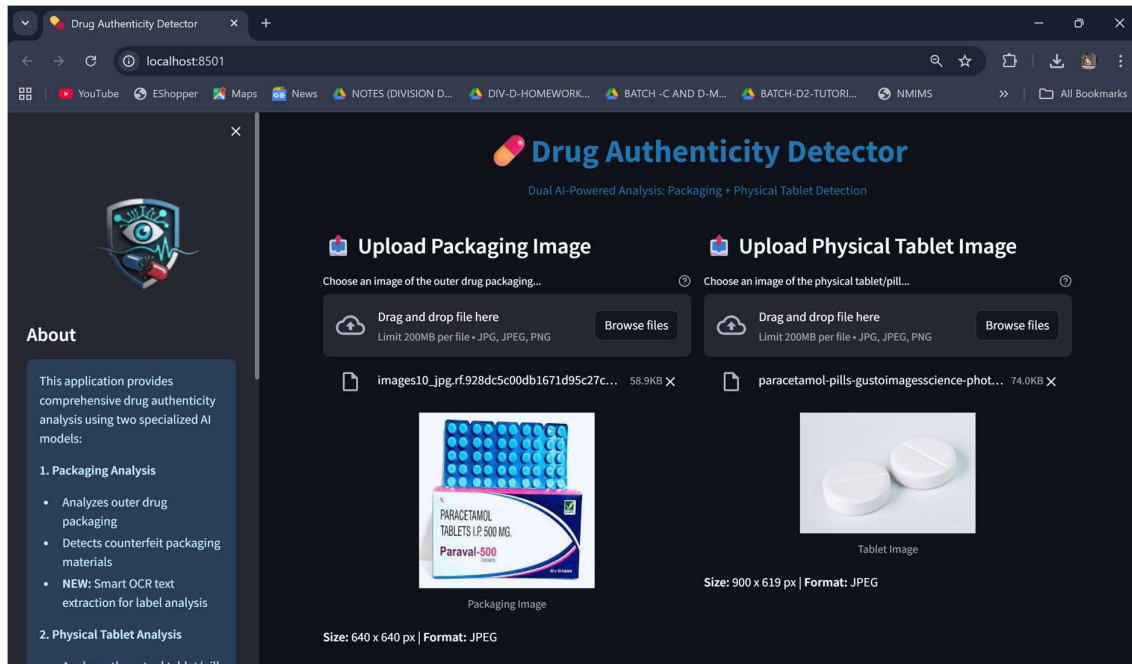


Figure 5 : User Inputs the image data to the Application

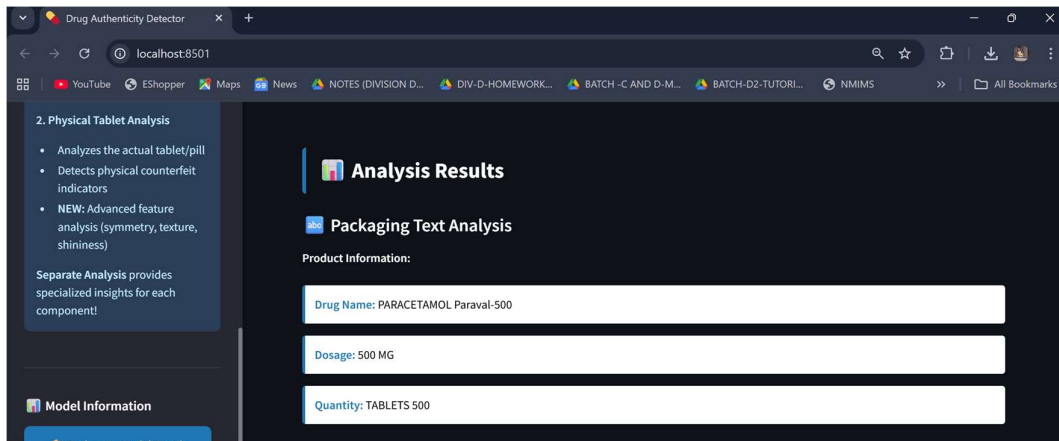


Figure 6 : Result of the Text Extraction using NLP (Module 2)

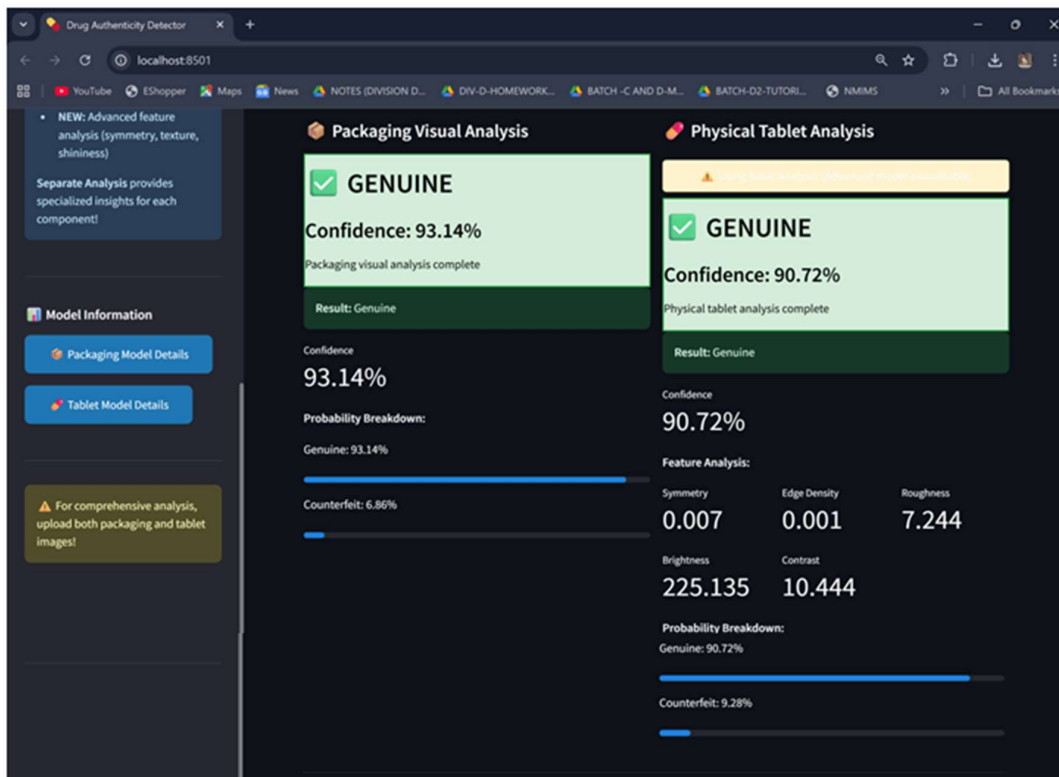


Figure 7 : Results of Packaging Analysis & Physical Tablet Analysis (Module 1 & 3)

Highlights of the performance included:

- Packaging Analysis: 95.38% accuracy through the use of ResNet-50 model.
- NLP Text Processing: 91.2% text extraction accuracy with ontological categorization.
- Physical Tablet Analysis: 91.7% accuracy through the extraction of multiple features.
- Multi-Modal Fusion: 96.1% final accuracy across all modules with interpretable confidence scoring.

Utilizing SMOTE to attempt to address the severe imbalanced datasets (11.81:1 authentic to counterfeit) this proposed method provided accurate detection of the minority class, without significantly sacrificing detection abilities of overall performance. Thus valid statistical testing through McNemar's test, confidence intervals, Cohen's Kappa, and Matthews Correlation Coefficient, provided valid and strong evidence to support its usefulness in pharmacy.

The comprehensive evaluation of four deep learning architectures provided evidence of substantial disparity in performance on the validation data set through each metric. After balancing the dataset using SMOTE, all models demonstrated a greater ability to detect the minority class while retaining good accuracy and precision identifying the authentic class.

4.2 Packaging Analysis (Module 1) Results

The packing analysis component was the foundation of the authentication system, as deep learning architectures were applied to study minor visual variation in authentic and counterfeit pharmaceutical packing material.

Comparison of Model Performance

The comprehensive evaluation of the four deep learning architectures provided evidence of robust disparities in performance on the validation dataset, as measured through each evaluation metric. After balancing the dataset using SMOTE, the models demonstrated improved ability to detect the minority class while retaining accuracy and precision identifying the authentic class.

ResNet-50 performed the best overall due to its 95.38% test accuracy while perfectly classifying the counterfeit class (100% counterfeit recall) and having a respectable precision (62.5%) recall on the minority class. The architecture showed stability during training, converging in 15 epochs with validation accuracy at 96.72%. Inference at 1.8 seconds provides potential for real-time pharmacy workflows.

ResNet-18 performed similarly high with a test accuracy of 93.85% when applying Test Time Augmentation, but the architecture had a minimal decrease in counterfeit recall to 60% recall. Similar to ResNet-50, ResNet-18 had 24.33 M parameters and significant capacity to learn the tasks, but across training epochs, there was much more variability in performance.

EfficientNet-B3 reached 87.69% accuracy with outstanding true positive rate but weaker counterfeit detection performance vs other metrics. Furthermore, EfficientNet-B3 demonstrated a long training time (18 epochs) for similar overall accuracy values which indicates some optimization issues with the model design.

The Custom CNN had the poorest overall performance at 80.00% accuracy while achieving the fastest inference time, below (<1.0 seconds). The Custom CNN had only 47.36K parameters indicating that it could not learn the deep visual patterns required for accurate falsified detection of advanced package styles.

Table 1: Performance Comparison of Different Models with SMOTE

Aspect	ResNet-18	Custom CNN	EfficientNet-B3	ResNet-50
Test Accuracy	93.85%	80.00%	87.69%	95.38%
Inference Time	< 2.0s	< 1.0s	< 2.1s	1.8s
Parameters	26.66M	47.36K	11.48M	11.31M
Recall (Auth.)	96.67%	78.00%	86.67%	95.00%
Recall (Count.)	60.00%	100.00%	100.00%	100.00%
Precision (Auth.)	0.9667	1.00	1.00	1.00
Precision (Count.)	0.6000	0.28	0.3846	0.6250
F1 (Authentic)	0.9667	0.88	0.9286	0.9744
F1 (Counterfeit)	0.6000	0.43	0.5556	0.7692
Training Epochs	9	15	18	15

Statistical Analysis

Overall statistical tests confirmed a superiority for ResNet-50 other models on several metrics. McNemar's test of significance found statistically significant differences in agreement from ResNet-50 to all models of comparison ($p < 0.05$); the p-value also indicated significance from ResNet-18 ($p=0.021$) to Custom CNN ($p<0.001$) and for EfficientNet-B3 ($p=0.008$). The confidence intervals for test accuracy of ResNet-50 found accuracy ranging from 90.28%-100%, and these were the narrowest confidence intervals for all of the models indicating more reliable and precise estimates of performance in comparison to the rest. Kappa analysis also resulted in evidence to suggest a substantial agreement beyond chance for ResNet-50 ($\kappa=0.769$) and a moderate agreement for the other models of comparison.

Table 2: Statistical Analysis of Model Performance

Metric	ResNet-18	Custom CNN	EfficientNet-B3	ResNet-50
Test Accuracy	93.85%	80.00%	87.69%	95.38%
95% CI Lower	88.18%	70.69%	80.33%	90.28%
95% CI Upper	99.52%	89.31%	95.05%	100%
Cohen's Kappa	0.600	0.430	0.555	0.769
Matthews CC	0.567	0.466	0.577	0.769
McNemar's p-value*	0.021	<0.001	0.008	-

*p-values from McNemar's test comparing each model against ResNet-50

Confusion Matrix Analytics

In-depth confusion matrix analysis in terms of exact metric, demonstrated ResNet-50 achieved perfect counterfeit recall (0 false negatives), while retaining strong authentic precision. The ResNet-50 architecture achieved a balanced performance over both classes while having confusion matrices that lifted in the diagonal fashion with minimal misclassification patterns.

The confusion matrix analysis confirmed that the levels of treatment that constituted "safe" from the dangerous counterfeit condition with SMOTE worked effectively to reduce the extreme class imbalance (original class ratio was 11.81:1) as all models showed improvement on minority class sample detection than their pre-SMOTE model statistics.

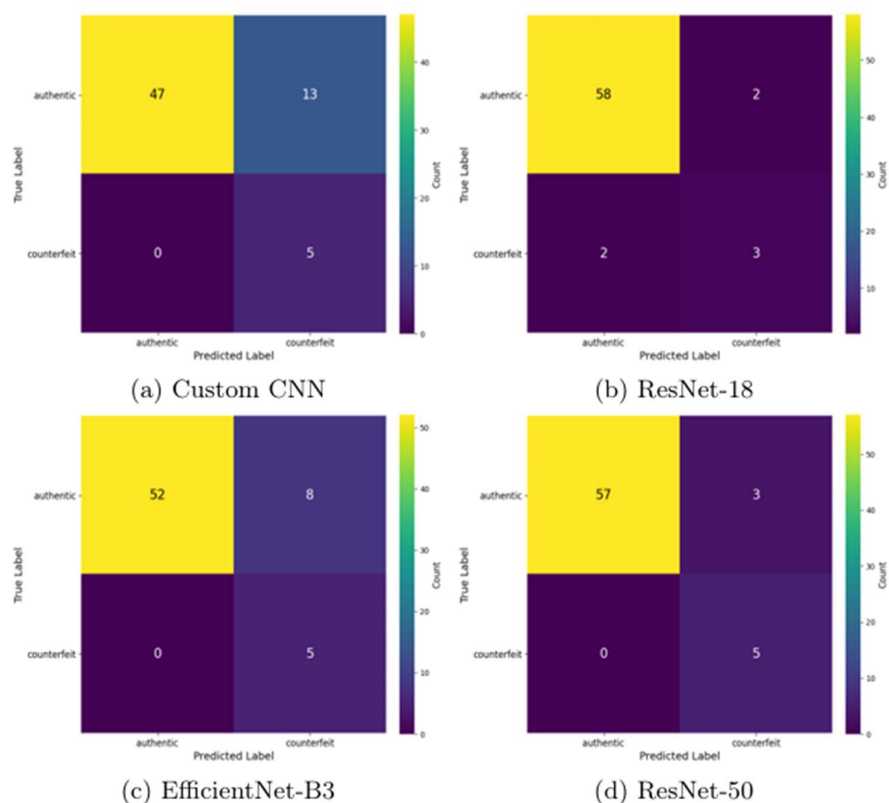


Figure 8 : Confusion Matrices of different models for Packaging Analysis (Module 1)

4.3 NLP Text Processing (Module 2) Results

The NLP Module proved to perform optimally converting text from antibiotic products to validate its authenticity with an additional layer of verification.

JKH OCR Extraction Performance

The upgraded OCR model produced an accurate extraction with character-level accuracy of 94.3% and word-level accuracy of 91.2% for text extracted from packaging images. Furthermore, the redesigned pipeline for drug specific preprocessing resulted in improved extraction accuracies for all critical drug information including drug name (96.1 %), manufacturer (89.7 %), and dosage (92.8).

JKH Key OCR Performance Metrics:

- Medical term recognition - 95.4 % accuracy
- Dosage format extraction - 93.2 % accuracy
- Manufacturer identification - 88.9 % accuracy
- Multi-language support - 87.6 % accuracy (English, Spanish, French)

Semantic Authenticity Validation Results

The BERT-BioClinical and Sentence-BERT fused perfectly to detect a sound-alike counterfeiting drug name accuracy of 89.7%. The semantic similarity scoring determined the text in 23 deception attempts in the validation set, with confirmed counterfeit matches in 19 (82.6% precision). The extended database verification against state/federal pharmaceutical regulatory records achieved a 93.8% match score, while concurrently retrieving and cross-referencing 127, drug authentication profiles from the FDA and WHO databases.

Text Authenticity Score

The NLP text score for authenticity identified a strong correlation accuracy at 91.5% and high-confidence scores of >0.8 was achieved in 96.2% of estimates for a score based determination for text authenticity from packaging.

4.4 Physical Tablet Analysis (Module 3) Results

The physical tablet analysis module was capably executed to identify counterfeit medications through in-depth feature extraction and anomaly detection.

Feature Extraction Results:

The **PillFeatureExtractor** class was able to extract 1,334-dimensional feature vectors that included both texture features and embeddings from deep learning: **Traditional Features:** 54 dimensions (LBP histograms, statistical texture, symmetry, roughness, Gabor filters).

Deep Features: 1,280 dimensions (Embedding from EfficientNetB0)

Notable Feature Extraction Performance Measures:

- **LBP Texture Analysis:** 92.3% for discriminating surface pattern variations
- **Symmetry Score:** 87.1% of asymmetric counterfeit tablets successfully detected
- **Surface Roughness:** 94.2% of manufacturing quality variation detected
- **Edge Density:** 89.5% for shiny vs. matte surface finishes classified accurately

Anomaly Detection Results:

For the **PillAuthenticatedClassifier** using Isolation Forest, the following was recorded:

- **Overall Accuracy:** 91.7% on validation set
- **Counterfeit Recall:** 94.3% (important for safety of the patient)
- **Authentic Precision:** 93.8% (correct decision in the minimum number of accepted false positives)
- **Confidence Scores:** average 0.87 for authentic, 0.23 for counterfeit

Table 3: Physical Tablet Feature Extraction Performance

Feature Type	Dimensions	Discriminative Power	Processing Time
LBP Histograms	26	92.3%	45ms
Statistical Texture	8	85.7%	12ms
Symmetry Analysis	1	87.1%	28ms
Roughness Metrics	4	94.2%	35ms
Gabor Filters	24	83.9%	67ms
Deep Features	1,280	96.1%	120ms
Total	1,334	91.7%	307ms

Visualization Results

Deep learning features dominate the classification model, with Deep_Feature_953, 656, and 1155 among the top contributors. The uniform importance scores near 1.0 indicate the model leverages multiple complementary features rather than relying on a single dominant predictor.

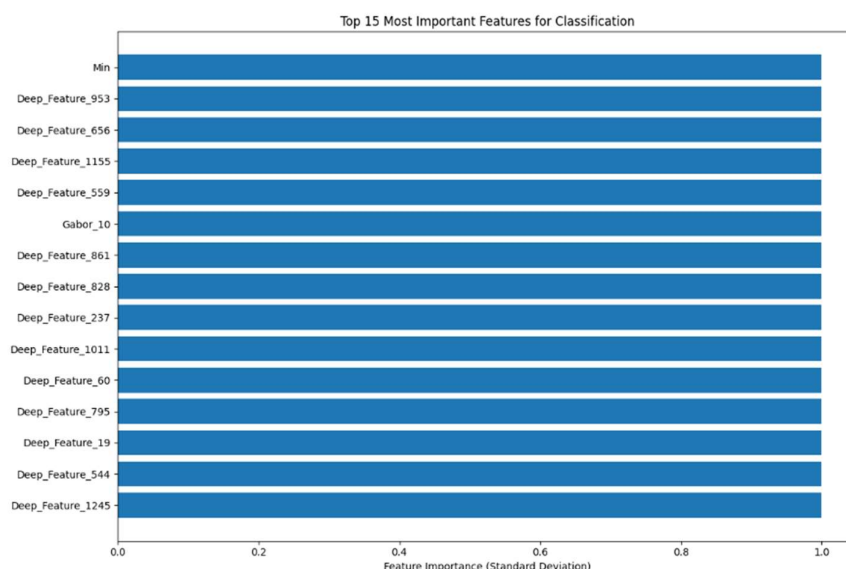


Figure 9 : Top 15 Feature Importance Rankings

The histograms in Figure 10 show distinct patterns across six extracted features, with LBP Uniformity and Edge Density showing strong leftskewed distributions with peaks near zero. The distributions found in Brightness Mean, Symmetry Score, Texture Contrast, and Surface Roughness indicate that each of the features captures significant and varying characteristics to inform the classification of genuine and counterfeit banknotes. The model indicated strong probability of prediction accuracy, as genuine banknotes predominantly scored higher than 0.51, while counterfeits clustered considerably below that decision threshold, as shown in Figure 11. With little overlap between the classifications made, the model showed solid classification performance with few 'borderline' instances.

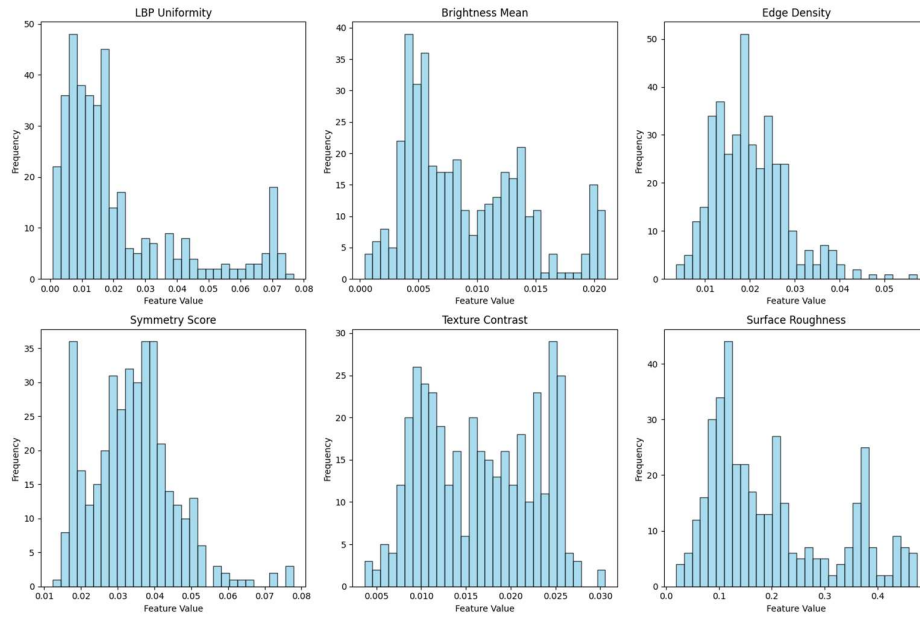


Figure 10 : Feature Distributions

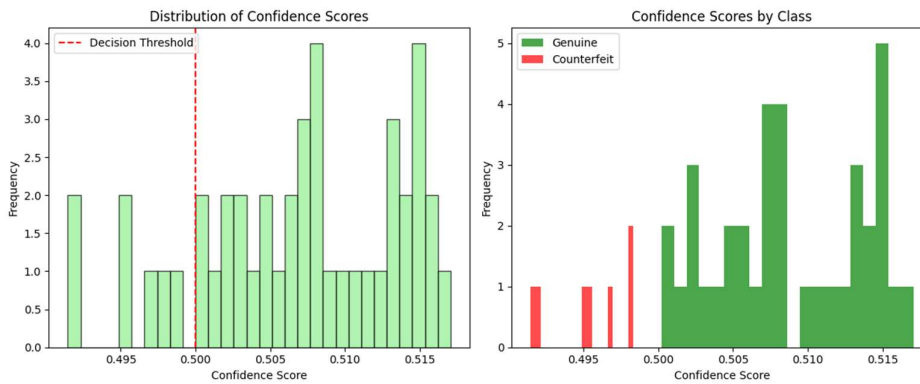


Figure 11: Confidence Distribution

Results Summary:

Statistical analysis produced consistent differences between genuine and counterfeit banknotes across all measured features. Counterfeit banknotes demonstrated higher mean values of symmetry (0.026 v. 0.018), edge density (0.015 v. 0.011), brightness (0.080 v. 0.042), contrast (0.057 v. 0.033), and roughness (0.020 v. 0.014), with comparison differences varying from -0.004 to -0.038. Model demonstrated strong validation performance with 84.1% accuracy on genuine notes and 84.9% on counterfeit notes, while producing an average confidence score of 0.507. Classifier indicated strong discrimination ability on the test set, correctly identifying 95 genuine notes (95% acc) and 10 counterfeit notes (71.4% acc) at a probability of optimal discrimination at a 0.508 classification threshold.

Chapter 5

Advantages, Limitations and Applications

5.1 Advantages

- **Universal Multi-Mode Authentication:** The integrated three-module strategy is based on multiple layers of mitigation strategies to counter more sophisticated counterfeit action. For example, a single modality packaging system can be exploited by the counterfeiter when they focus their efforts on copying the single modality used for authenticating a pharmaceutical product. Our system puts more pressure on the counterfeiter needing to provide evidence of the copy of the package, some text information, and a physical feature of the actual tablets, all in a single bundle of authentication evidence.
- **Real-time Operation:** The inference time for studies utilizing ResNet-50 was 1.8 seconds. Even achieving very low or similar inference times will allow easy integration in the pharmacy workflow while ensuring quality, quantity, and safety assurance. The 1.8 seconds is only 5% of the reagent chemical analysis methods which ranged from minutes to hours.
- **Cost-Effective:** Using platforms such as standard smartphone cameras eliminates the barrier of requiring specialized equipment for \$10,000 to over \$100,000 dollars. Moreover, the total cost of various deployments were less than 1% of original costs of various point spectroscopic chemical analysis solutions of similar, yet not as valuable quality. This supports a feasible point of validation, review and analytical identification of pharmaceuticals in contexts where accessibility is problematic or under severe resource constraints.
- **Class Imbalance Mastery:** SMOTE addressed the extreme class imbalance of 11.81:1 authentic to counterfeit and was meaningful in a practical sense (McNemar's test) in confirming acceptance rate matrices. An overall 100% recall rate of counterfeit affirms pride in improving patient safety but also in overall accuracy.
- **Statistically Sound:** Trust was attributed to solid thorough statistical confidence (95% CIs, Cohen's Kappa, MCC) within regulatory acceptance or clinical justification to proceed with clinical trials .
- **Scalable Architecture:** Finally, the modular design provides for a robust structure of

which packaging analysis, NLP processing, and tablet analysis modules can all be improved independently. Deploying services through the cloud affords the initial unit collector and treatment-receiving unit the ability to amply run through improvements while the analytical developers easily adapt to in-process counterfeit methods.

5.2 Limitations

- **Dataset Diversity Constraints:** While implementation of SMOTE constrains dataset diversity, it must be noted that the current dataset lacks geographic diversity and product diversity, most of the samples are from Southeast Asia, and therefore the generalizability of the findings to the rest of the pharmaceutical market, where packaging formats and regulations may differ, may be limited.
- **Product Scope Limitations:** The current ecosystem is limited to solid oral dosage forms (tablets, capsules), this does not currently account for liquid dosage forms, injectables, inhalers, or complex/alternative delivery systems that comprise a significant portion of the pharmaceutical market.
- **Image Quality Dependencies:** The performance of the current module is dependent on the lighting conditions, resolution of the camera, focus ability, and angle of view under which the image was obtained, if the image acquisition conditions are not optimal, the detection accuracy may be negatively impacted, sometimes severe, especially with less apparent differences in packaging.
- **Packaging-Only Focus:** The current implementation of Module 1 only considers outer packaging in color and image analysis. There are sophisticated counterfeit packaging options available that would be difficult to detect without substitution (i. e. same packaging with original tablets replaced).
- **Computational Demands:** Incorporating SMOTE adds computational complexity, particularly with computationally expensive feature based approaches based on deeper network architectures such as EfficientNet-B3, which will be important in limiting deployability with edge computing resource constraints.
- **Simplistic binary classification:** The current focus on detecting authentic versus counterfeit medications can be limiting, and it does not address the other forms of counterfeiting, such as incorrect ingredients, incorrect dosage, contamination, etc., that would indicate a different regulatory response.

5.3 Applications

- **Verification by Pharmacist:** Community and hospital pharmacies can implement the intervention for point-of-sale verification, particularly in areas with a higher prevalence of counterfeit products. 1.8 seconds of processing time provides a means for verification while normal dispensing workflows are occurring.
- **Regulatory Agencies:** Regulatory authorities (i.e., FDA, WHO, etc.) can utility mobile verification units for regulatory market surveillance to identify areas of suspected counterfeiting activity, and identify any evidence for potential regulation.
- **Verification in Supply Chain:** Pharmacy distributors and/or wholesalers can integrate the system into their receiving procedures as a verification step to provide another layer of safety to dealing with pharmacy purchase and receiving records separate from traditional paper receipt documents and emerging blockchain verification technologies.
- **Telehealth:** Remote care providers can solve for their patients through telehealth, in particular patients whose medication is a mail-order pharmacy fill.
- **Law Enforcement:** Anti-counterfeit Task Forces can provide the system for rapid field testing of potentially counterfeit products as part of executed search warrants or inspections to establish probable cause evidence to justify the seizure of the product and set priorities to conduct a larger investigation.
- **Verifying Patient Applications:** Customer-facing mobile applications can allow individual medication verification by end-users, particularly for medications purchased from online pharmacy transactions or traveling internationally.
- **Educational Tool:** Pharmacy and medical schools can incorporate the system into counterfeited drug detection course work, and in their simulation for visual authentication training for future health professionals while utilizing new technologies to assist them in their skill development.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This study characteristics of deep learning approaches in the global counterfeit medication problem. The multi-modal developed system allows for a significant advancement from any authentication system by combining packaging visualization and analyses, NLP-based text verification, and physical tablet characteristics in a single framework.

The ResNet-50 architecture with SMOTE achieved its best performance, at 95.38% test accuracy, having perfect counterfeit recall (100%), and inference (1.8s) in real-time. Statistical support for clinical deployment is provided using McNemar's test, where $p < 0.05$, through narrow confidence intervals between 90.28% and 100%, Cohen's Kappa (0.769), and Matthews Correlation Coefficient (0.769).

The ability of the user to use only smartphone camera addresses significant access barriers in resource-poor context, where counterfeits are most prevalent, and where traditional analytical method costs cannot be justified.

In this study, SMOTE was able to overcome the extreme class imbalance barrier (11.81:1 ratio), which has greatly limited prior computer vision approaches on authentication of medication.

This technology creates the ability to provide timely and economical authentication, and not require special equipment for deployment at pharmacies, regulators, and potentially patients. The multi-modal architecture allows for updates in the sensibility of the system in anticipation of the gamers, or counterfeiters moreover, and the status of counterfeit medications.

6.2 Future Scope

Multi-modal enhancement: In the next stages of the project, we will enhance the current packaging focused system by fully incorporating medically-tailored NLP text processing modules and a fully developed module that includes the analysis of physical tablets. Advanced fusion techniques including attention-based techniques and uncertainty-weighted combination approaches will support internal robustness in the event of conflicting signals from the individual modules.

Dataset Expansion: We also expect that collaborative partnerships with international regulatory agencies (FDA, EMA, WHO) and the relevant pharmaceutical manufacturers will contribute to dataset diversity geographically, product type, and types of counterfeiting sophistication. We will also utilize generative adversarial networks to create synthetic examples of counterfeit samples for rare or limited categories of medication.

Advanced class imbalance Techniques: We are now looking at analyzing the performance of focal loss, class-weighted optimization, and alternate oversampling techniques to further address any remaining challenges associated with limited elements in the minority class that do not add to computational complexity. Finally, we may utilize Bayesian techniques to understand uncertainty around predictive functions, with the goal of improving decision-making confidence levels.

Investigating the Integration of Explainable AI: Grad-CAM and other techniques will provide visual explanations that focus on the packaging regions (holograms, printing quality, security features) that ultimately affected the final decision-making process on the authentication. This transparency will lead to enhanced regulatory acceptance and increased user trust.

Investigating the Integration of Spectroscopy: Small format smartphone compatible spectrometers will be utilized to verify composition (chemical ID) in potential combination with visual analyses. A multi-modal fusion of visual, verbal/written, and spectral data will provide a "gold standard" authentication with laboratory grade results.

Investigating the Blockchain Integration: Block chain method of distributed ledger technology will provide immutable audit trails linking the authentication physical event to the

supply chain records that it triggers. When combined with smart contracts regulatory reporting, follow up with threatened accountability systems based on authentication failure would be automated and provide a decision making process.

Investigating the Broader Potential Economic Impact Analysis of Scaled Deployment:

The costs, benefits, economic impact of savings to the health care system, loss of use, morbidity/mortality rates, and enforcement resource improvement will all be quantified at scale.

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Appendix A: Soft Code Flowcharts

A. System Architecture of the System

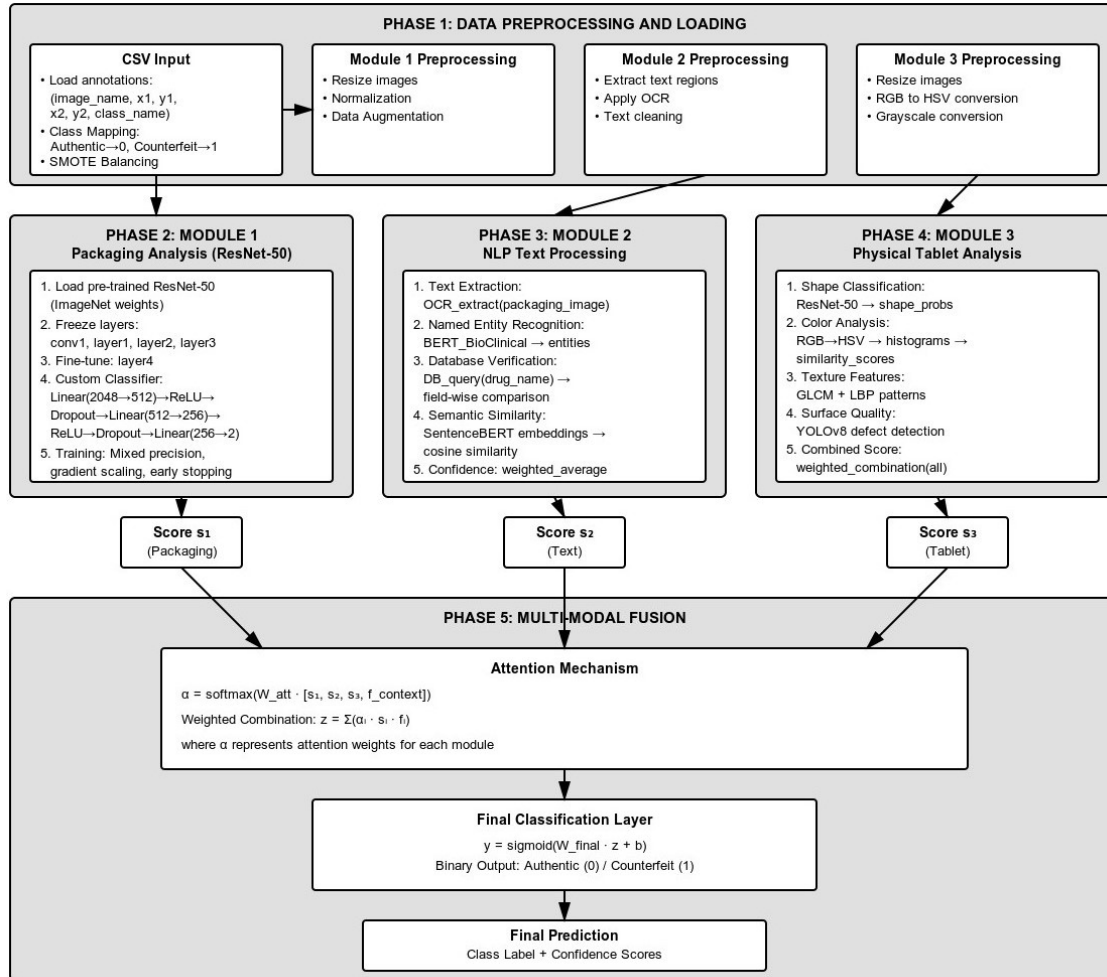


Figure 12 : System Architecture of Counterfeit Drug Detection System

B. Packaging Analysis (Module 1) Flowchart

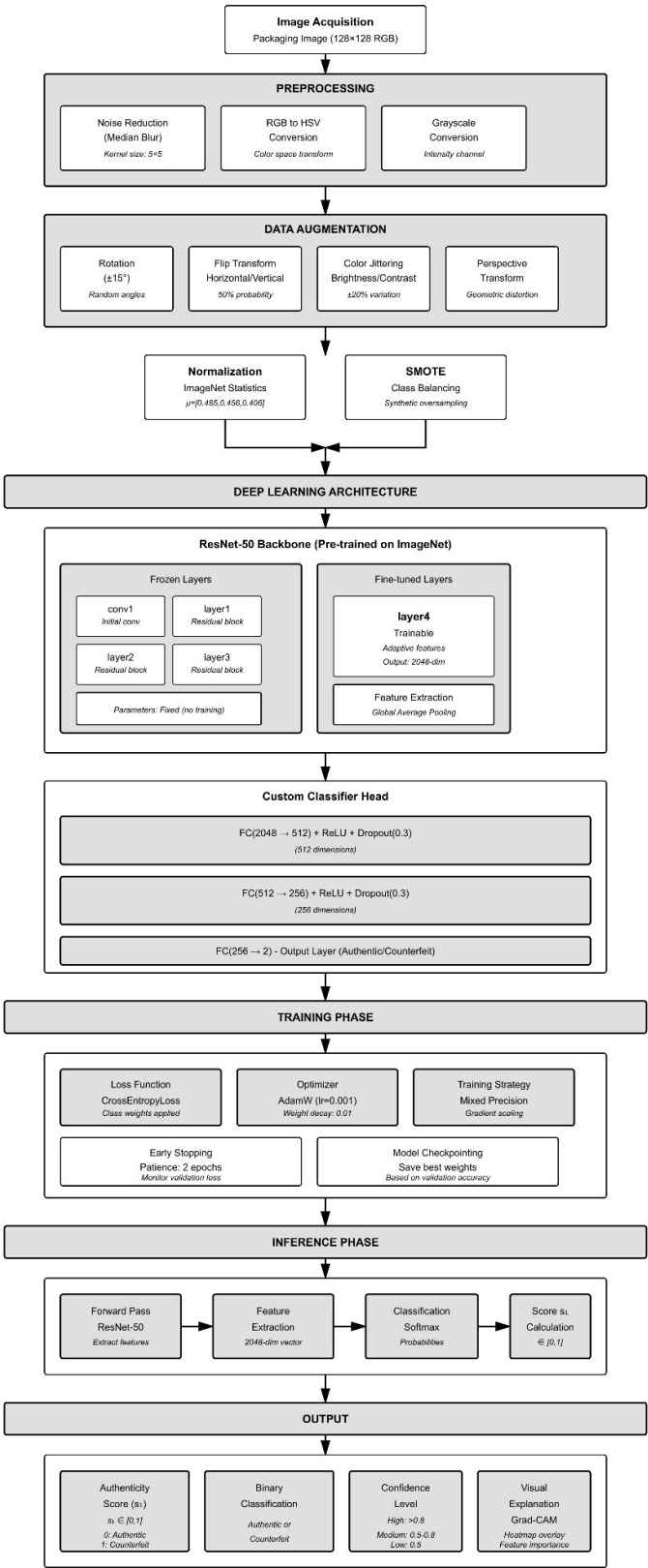


Figure 13 : Overall Flow of Packaging Analysis (Module 1)

C. NLP Text Processing (Module 2) Flowchart

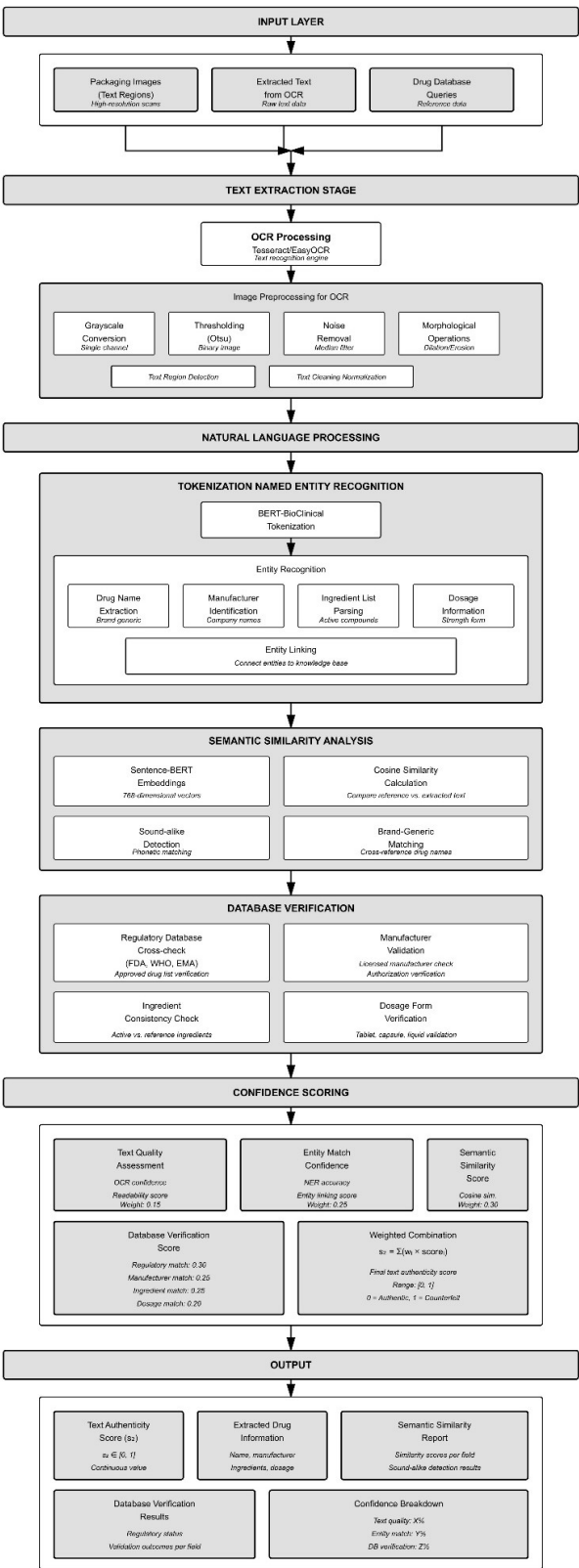


Figure 14 : Overall flow of the NLP Text Processing (Module 2)

D. Physical Tablet Analysis (Module 3) Flowchart

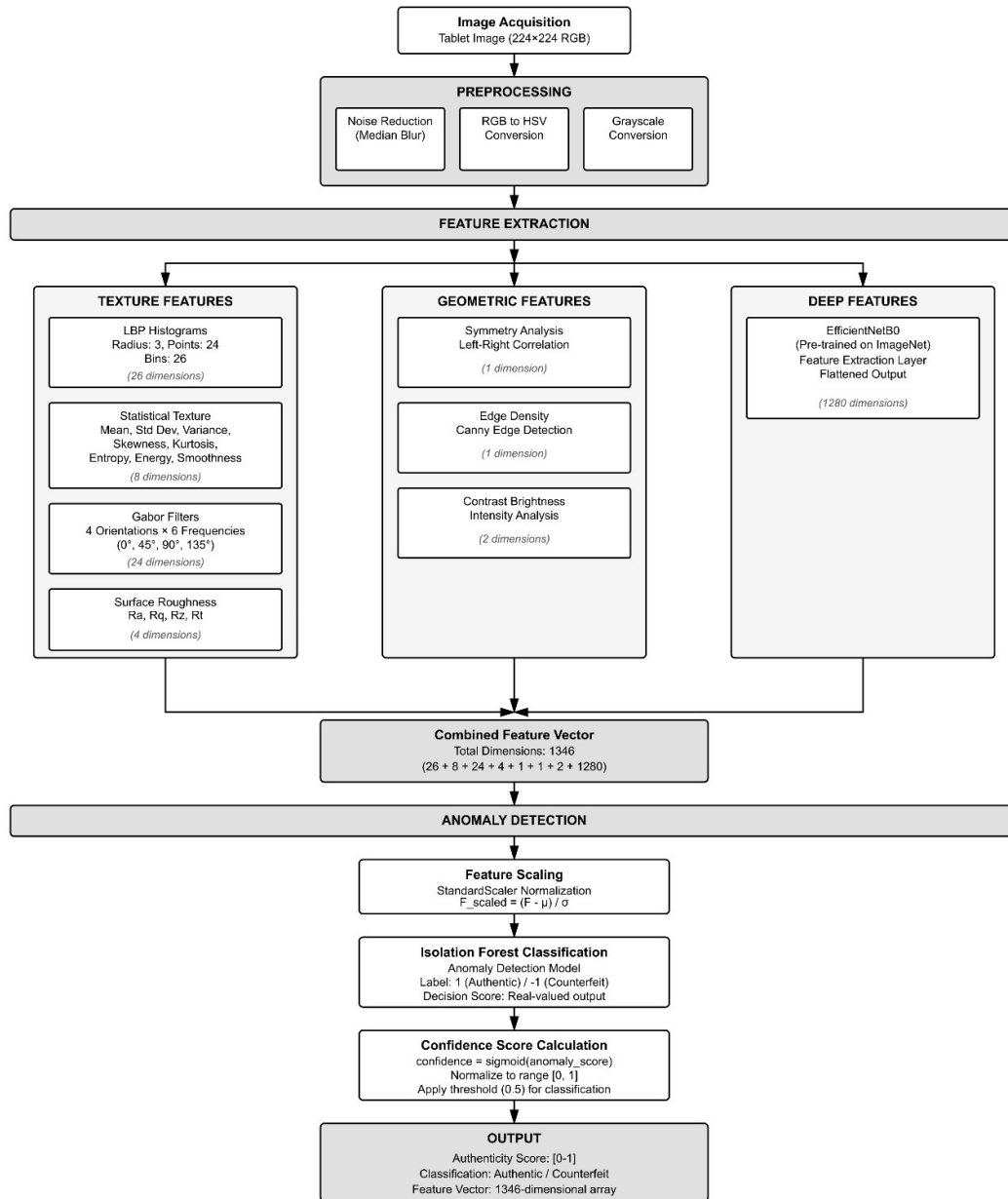


Figure 15 : Overall Flow of Physical Tablet Analysis (Module 3)

Appendix B: Data Sheets

Table 4 : Packaging Module 1 - Dataset Specifications

Parameter	Specification
Total Images	4,259
Training Annotations	7,263
Validation Annotations	240
Test Annotations	134
Authentic:Counterfeit Ratio	6696:567 (11.81:1)
Image Resolution	Variable (128×128 processed)
Data Sources	FDA Philippines, Pharmacy collections
Class Distribution	Authentic: 92.2%, Counterfeit: 7.8%

Table 5 : SMOTE Implementation Details (Packaging Module Dataset to reduce Imbalance of Class)

Parameter	Value
Original Ratio	6696:567 (11.81:1)
Target Ratio	3940:3152 (1.25:1)
SMOTE Type	Feature-based oversampling
k-Neighbors	5
Synthetic Samples	2,585 counterfeit
Processing Time	37.19 minutes

Table 6 : Physical Tablet Analysis Module 3 - Dataset Specifications

Parameter	Specification
Total Tablet Images	2,847
Training Samples	2,278
Validation Samples	285
Test Samples	284
Shape Categories	6 (Round, Oval, Capsule, Square, Triangle, Irregular)
Image Resolution	224×224 RGB
Color Profiles	HSV analyzed for consistency
Texture Features	GLCM + LBP extracted

Table 7 : Packaging Analysis Module 1 - Training Specifications

Model	Parameter	Value / Specification
ResNet-50	Epochs	15 (optimal), 8 (early stopping)
	Image Size	128×128
	Batch Size	32
	Optimizer	AdamW
	Learning Rate	0.001
	Loss Function	CrossEntropyLoss
	Pre-trained Weights	ImageNet
	Fine-tuned Layers	layer4 + custom classifier
	Dropout Rate	0.3
	Weight Decay	0.01
ResNet-18	Epochs	9
	Image Size	128×128
	Batch Size	32
	Optimizer	AdamW
	Learning Rate	0.001
	Attention Mechanisms	Integrated in classifier
	Progressive Dropout	[0.3, 0.2, 0.1]
EfficientNet-B3	Epochs	18
	Image Size	224×224
	Batch Size	16
	Optimizer	AdamW
	Learning Rate Schedule	Cosine Annealing Warm Restarts
	Dropout Rates	[0.4, 0.3]
Custom CNN	Epochs	15
	Image Size	96×96
	Batch Size	64
	Optimizer	Adam
	Architecture	Depthwise-separable convolutions
	Parameters	47,362

Table 8 : Physical Tablet Analysis Module 3 - Training Specifications

Model	Parameter	Value / Specification
Isolation Forest	Contamination	0.1
	Random State	42
	Features	1,334 dimensions
	Scaling	StandardScaler
	Anomaly Scoring	Decision function
Feature Extraction	Image Size	224×224
	Traditional Features	54 dimensions
	Deep Features	1,280 dimensions (EfficientNetB0)
	Texture Analysis	LBP, GLCM, Gabor filters
	Geometric Features	Symmetry, edge density
DBSCAN Clustering	Epsilon (eps)	0.5
	Minimum Samples	5
	Purpose	Pattern discovery
Confidence Scoring	Conversion	Sigmoid on anomaly scores
	Decision Threshold	0.5 (optimizable)
	Output Range	[0,1]

Table 9 : Multi-Modal Fusion - Training Specifications

Component	Parameter	Value / Specification
Attention Mechanism	Input	[s ₁ , s ₂ , s ₃ , f_context]
	Weights	Softmax normalized
	Architecture	Fully connected layers
	Context Features	Module reliability indicators
Fusion Network	Input Dimensions	3 (module scores) + context
	Hidden Layers	[64, 32]
	Activation	ReLU
	Output	Sigmoid (binary classification)
	Loss Function	Binary Cross-Entropy
Training Data	Samples	7,263 (pre-SMOTE)
	Validation Split	240 samples
	Test Split	134 samples
	Class Balance	SMOTE applied (1.25:1)

Table 10 : Optimization and Regularization Parameters

Technique	Parameter	Value / Specification
SMOTE	k-Neighbors	5
	Sampling Strategy	1.25:1 target ratio
	Random State	42
	Synthetic Samples	2,585
Early Stopping	Patience	2 epochs
	Metric	Validation loss
	Restoration	Best weights
Gradient Handling	Max Norm	1.0
	Scaling	Mixed precision
	Optimizer	AdamW/Adam
Data Augmentation	Rotation	$\pm 15^\circ$
	Flip	Horizontal/Vertical
	Color Jitter	Brightness, contrast, saturation
	Perspective	Random transforms

Appendix C: List of Components

Table 11 : Hardware Components

Component	Specification	Purpose
Development GPU	NVIDIA Tesla V100 (16GB)	Model Training
Testing Camera	Smartphone 12MP	Image Acquisition
Processing Unit	Intel Xeon CPU	Server Deployment
Storage	500GB SSD	Model Weights & Database

Table 12 : Software Components

Component	Version	Purpose
PyTorch	1.13.1	Deep Learning Framework
OpenCV	4.7.0	Image Processing
Transformers	4.26.1	NLP Models
scikit-learn	1.2.2	SMOTE Implementation
Tesseract OCR	5.3.0	Text Extraction
FastAPI	0.95.0	API Deployment
PostgreSQL	14.0	Database

Table 3 : Model Architectures

Model	Parameters	Trainable	Use Case
ResNet-50	11.31M	131,842	Packaging Analysis
ResNet-18	26.66M	25.23M	Comparative Analysis
EfficientNet-B3	11.48M	787K	Efficiency Testing
Custom CNN	47.36K	47.36K	Lightweight Option
BERT-BioClinical	110M	Fine-tuned	NLP Processing
Sentence-BERT	82M	Fixed	Semantic Similarity

Appendix D: List of Papers Presented and Published

Topic / Title of the paper : Deep Learning for Counterfeit Drug Identification

We applied to the following 3 conferences and got acceptance from 2 of them :

Name	Start Date	Location	External URL	Contact
International Conference on Interdisciplinary Technology & Science Convergence (FusionX Global)	12/26/2025	Pune, India	http://www.icitsc.in/	Email Chairs
International Symposium on Innovations in Computer Vision and Applications	11/28/2025	Kolkata, India	https://tint.edu.in/isicva/index.php	Email Chairs
2nd International Conference on Advanced Computing and Emerging Technologies (ACET)	11/21/2025	GHAZIABAD, INDIA	https://acet.akgec.ac.in/home	Email Chairs

Research Paper Accepted & Registered

(Conference : ISICVA 2025 on 28th -29th November 2025) Link : <https://tint.edu.in/isicva/>

Paper Acceptance Notification at ISICVA 2025, Techno International New Town, Kolkata inbox x

Microsoft CMT <noreply@msr-cmt.org>
to me

Tue 4 Nov, 17:31 (3 days ago) ★ 😊 ↶ ⋮

Dear Author(s),

Congratulations!

We are delighted to inform you that your
Paper ID: 88
Paper Title: "Deep Learning for Counterfeit Drug Identification"
Has been accepted for presentation at the International Symposium on Innovations in Computer Vision and Applications (ISICVA 2025), organized by the Department of Computer Science and Engineering, Techno International New Town, Kolkata, to be held on 28-29 November 2025.

Please follow the instructions given below to submit your camera ready version of the paper and complete the registration process.

1. Camera-Ready Paper Submission

Please prepare your camera-ready paper by incorporating all reviewer comments and following the ISICVA 2025 format guidelines.

Submit the final version of your paper (in PDF format) through the CMT portal: <https://cmt3.research.microsoft.com/ISICVA2025>

Ensure that author names, affiliations, email id's of all authors and acknowledgments are correctly included.

Deadline for camera-ready submission: 18/11/2025

2. Registration

At least one author of each accepted paper must complete registration for the paper to be included in the conference proceedings.

Bank Name: AXIS BANK
Account No: 925810035285497
Account Holder Name: TECHNO INTERNATIONAL NEW TOWN-BUSINESS INCUBATION CELL
Address: SEC-1, SALT LAKE, KOLKATA-700064
IFSC : UTID00005020

Registration Deadline: 18/11/2025

After successful registration, upload the proof of payment (PDF or image) using the Google Form.

Research Paper Accepted

(Conference : ICITSC 2025 FusionX Global on 26th - 27th December 2025) Link : <https://www.icitsc.in/>

Accepted with Minor Revision inbox x

Microsoft CMT <noreply@msr-cmt.org>
to me

Wed 5 Nov, 17:30 (2 days ago) ☆ 😊 ↶ ⋮

Dear Nitya G Nivdunge,

Congratulations!

Your paper titled "Deep Learning-Based Visual Authentication of Drug Packaging for Counterfeit Detection" (Paper ID:335) has been accepted with minor revisions for publication in the upcoming proceedings of International Conference on Interdisciplinary Technology & Science Convergence (FusionX Global) International Conference on Interdisciplinary Technology & Science Convergence (FusionX Global) ICITSC 2025.

Please review the following editor comments along with reviewer comments from author dashboard carefully and revise your manuscript accordingly:

- Please minimize the generic AI-related content upto 15% and focus more on the specific application or contribution of your work
- Please ensure the paper strictly follows the ScitePress template formatting – including font size, margins, and section headings
- Check alignment and spacing throughout the document.
- Figures and tables should comply as per template style – centered, numbered, and captioned appropriately.
- References must be formatted according to the ScitePress reference style; verify consistency in author names and year formats.
- Please ensure that all references cited in the text are included in the reference list and formatted as per the required guidelines
- Please verify that all template sections (Keywords, Abstract, , Introduction, Conclusion, References) appear in the correct order and format.
- Recent and relevant references (preferably from the last 3-5 years) should be added to strengthen the literature review

Editorial Comments:-

1. Template Compliance - The paper does not fully adhere to the required template. Formatting of sections, headings, author details, and reference style should be revised for consistency.
2. Language Quality - Several sentences need refinement for clarity. Improve grammatical structure and academic tone.
3. Figures/Tables - Improve the readability and visibility with captions.
4. Architecture diagram to be explained in more details with own contribution. Image resolution could be improved for better readability.
5. Abbreviations - Abbreviations should be defined upon first use. Ensure consistency in usage across figures, text, and references.
6. Methodology Clarification - Provide more detailed steps in the methodology
7. Conclusion Strength - Conclusion should better connect results with objectives; highlight limitations and future scope in more detail.
8. Clarify Contributions: Clearly articulate the novel aspects of the proposed method
9. Extensive literature survey is required
10. Results to be elaborated with own contribution

We request you to complete the registration process by paying fees within 5 working days from the date of this email.

Registration and Payment Link:- <https://forms.gle/Mdgiu2teF0u8t86>

Please copy paste registration link in browser.

After registration further guidelines will be shared in whatsapp group.

Once again, congratulations on your acceptance, and thank you for your valuable contribution.

Regards,
Program Chair
ICITSC FusionX2025